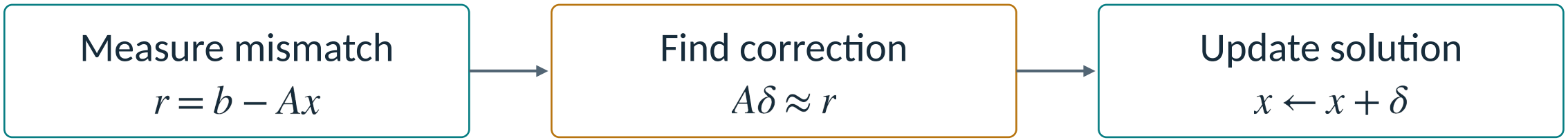
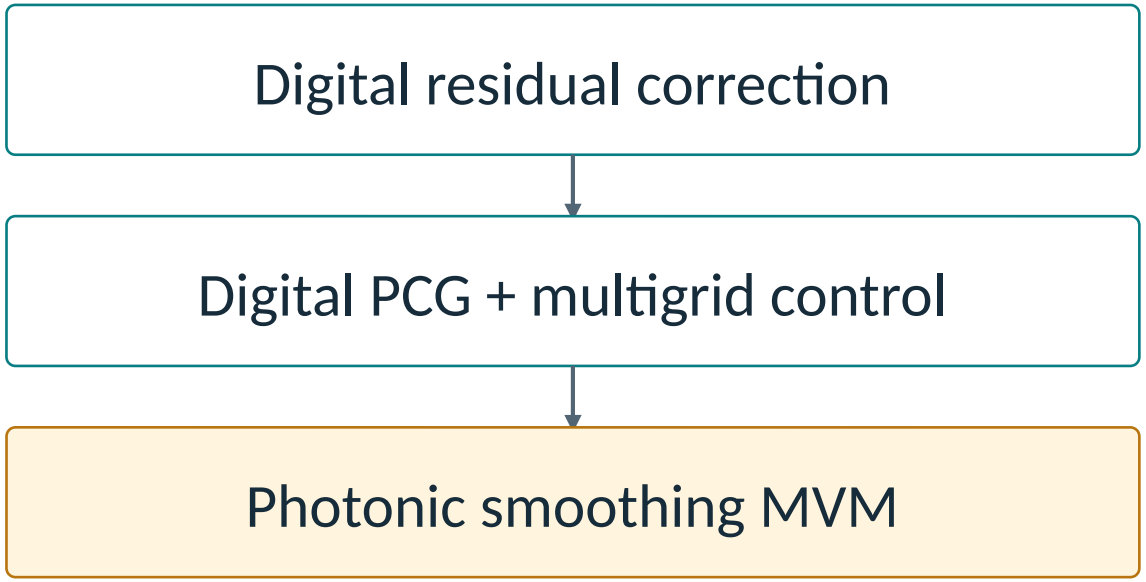
A digital iterative solver with photonic matrix–vector products

Fixed matrices and changing correction vectors solve Ax = b



The paper assigns smoothing products to optics

Numerical structure adopted here



Current project

Noisy 2D emulation
+
ideal 1D demonstrator

Hardware diagrams show functional assignments.

Research question: can approximate smoothing reduce digital work while reliable residuals retain accuracy?

Three implementations, different settings

	Lee Poisson experiment	main.py	Browser demonstrator
Role	Research foundation	Noisy numerical evidence	Explanatory model
Problem	2D capacitor	64 × 64 = 4096 unknowns	1D, 64 unknowns
AMG levels	2	4096 → 704 in this run	64 → 22 → 8 → 3
Hybrid sweeps	4 pre / 0 post	4 pre / 0 post	3 pre / 3 post
MVM behavior	Physical measurements	Input quantization + noise	Ideal floating point

Main explanation: browser’s 1D model. Research evidence: main.py’s separate 2D run.

The unknown vector stores sampled solution values

Ideal 1D teaching model

$$-\frac{d^2u}{dt^2} = 1$$

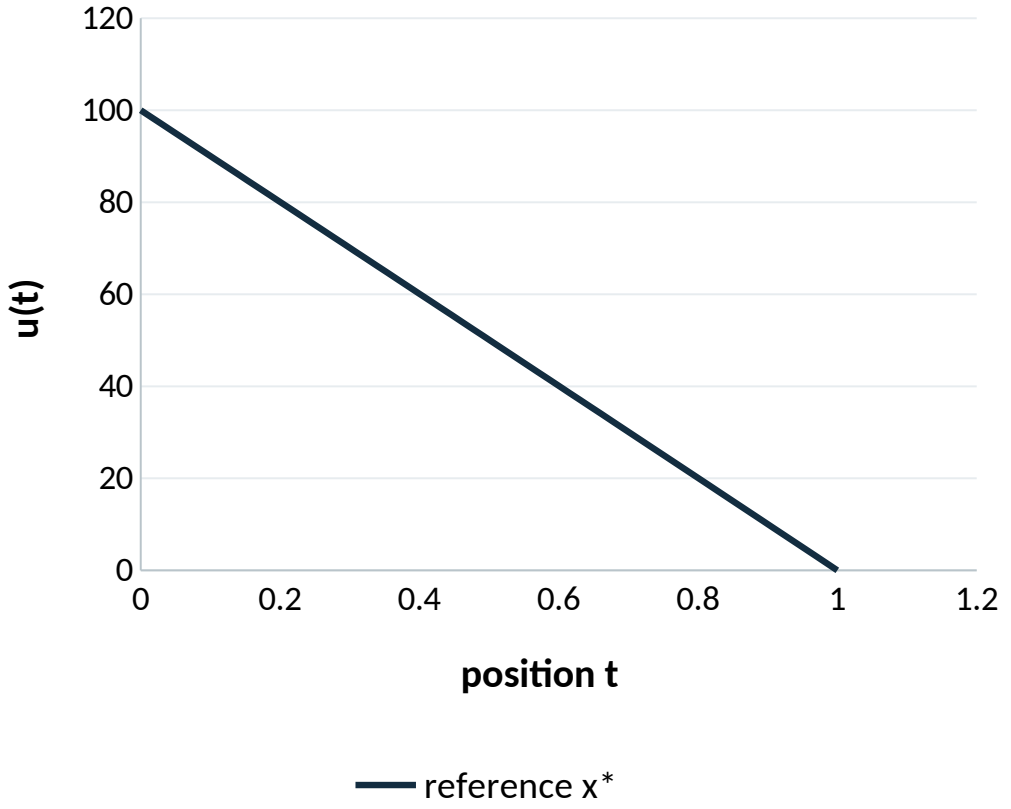
$$u(0) = 100, u(1) = 0$$

$$\frac{2x_i - x_{i-1} - x_{i+1}}{h^2} = 1$$

$$h = 1/65, x \in \mathbb{R}^{64}$$

$$A_0 = \frac{1}{h^2}\text{tridiag}(-1, 2, -1)$$

$$b = [422501, 1, \ldots, 1]^T$$



x = [u(t₁), …, u(t₆₄)]ᵀ. Iteration improves an equilibrium estimate; it is not physical time.

The residual becomes the next correction problem

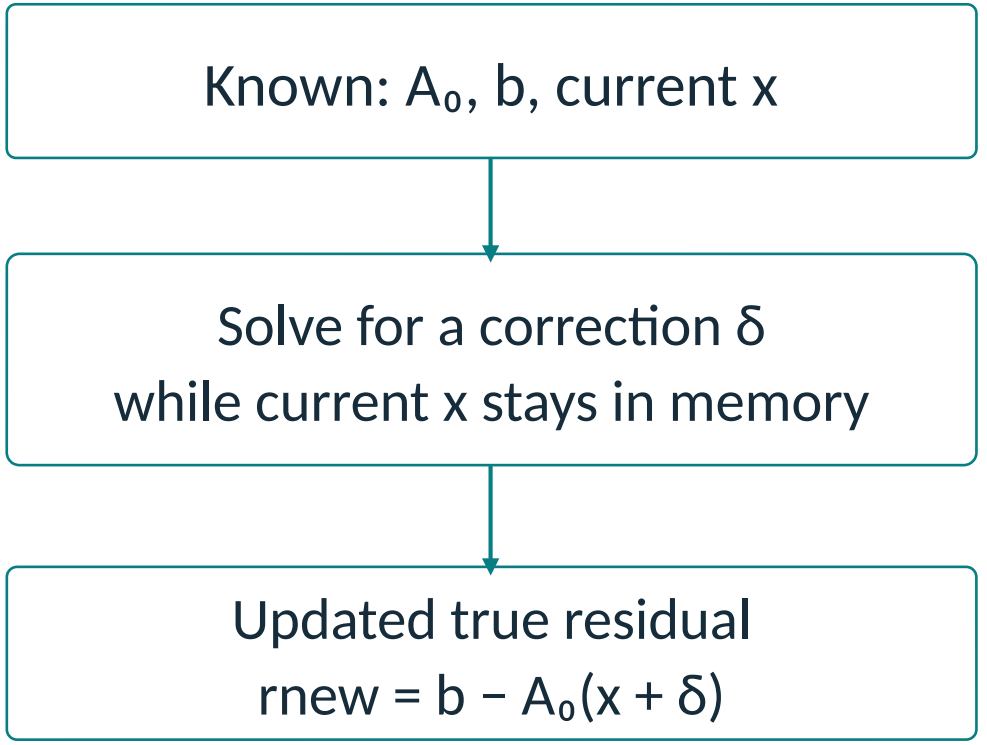
True residual check

$$r = b - A_0 x$$

$$A_0(x^\star - x) = r$$

Find δ : $A_0\delta \approx r$

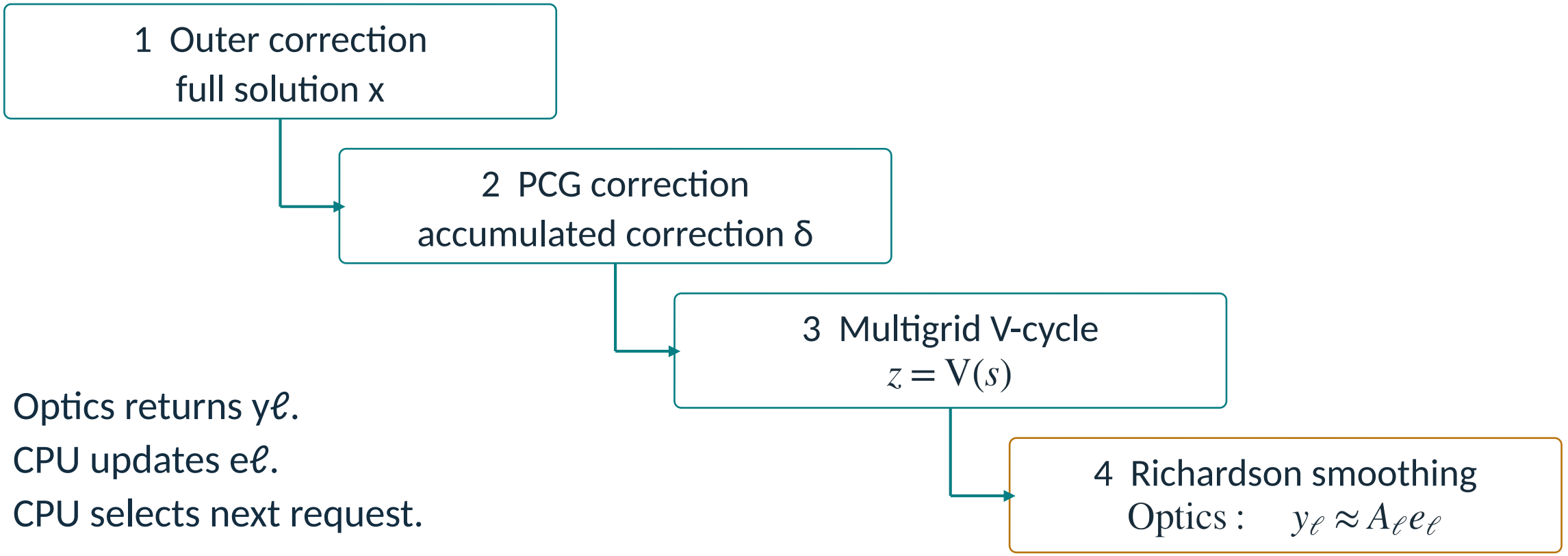
Then $x_{\text{new}} = x + \delta$



The exact solution x* is only a reference for evaluation. The algorithm uses the computable residual.

One solver map connects all four routines

Read downward for calls; read upward for returned values



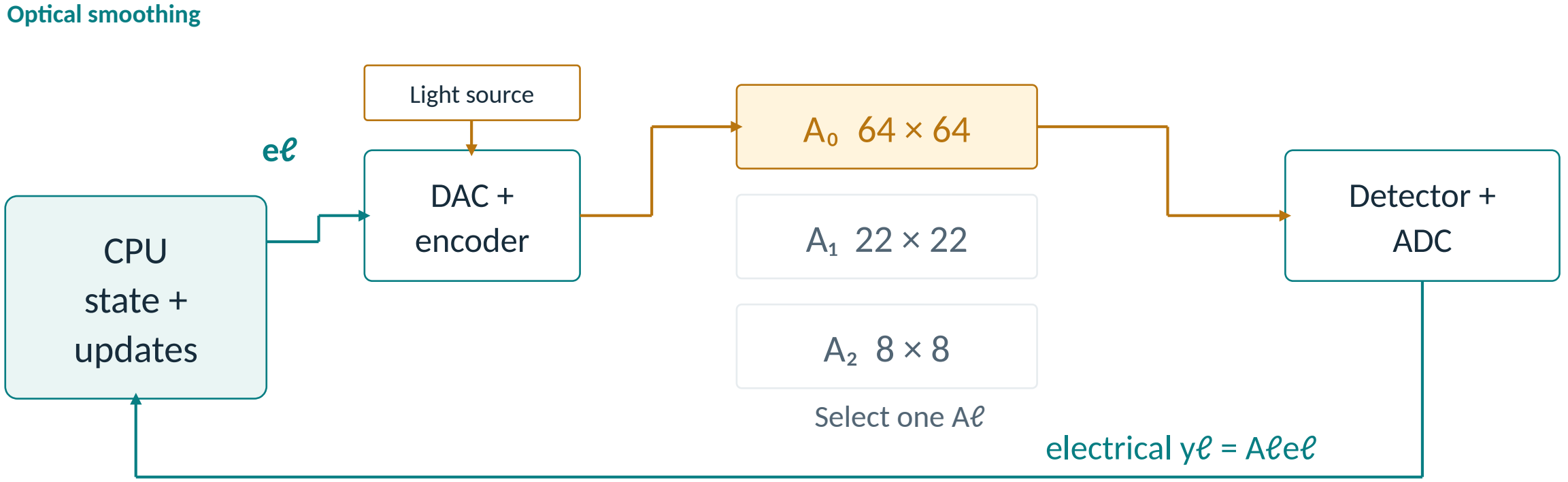
Optics returns y_ℓ.

CPU updates e_ℓ.

CPU selects next request.

Changing vectors: x → δ → s → e_ℓ. Fixed operators: A_ℓ, P_ℓ, R_ℓ.

The CPU closes every optical smoothing loop



$$\text{CPU update}: \quad e_\ell \leftarrow e_\ell + \omega_\ell (g_\ell - y_\ell)$$

Amber: optical product. Teal: electrical data and CPU work. The full solution x is not the encoded vector.

A real smoothing request changes the local vector

Optical smoothing

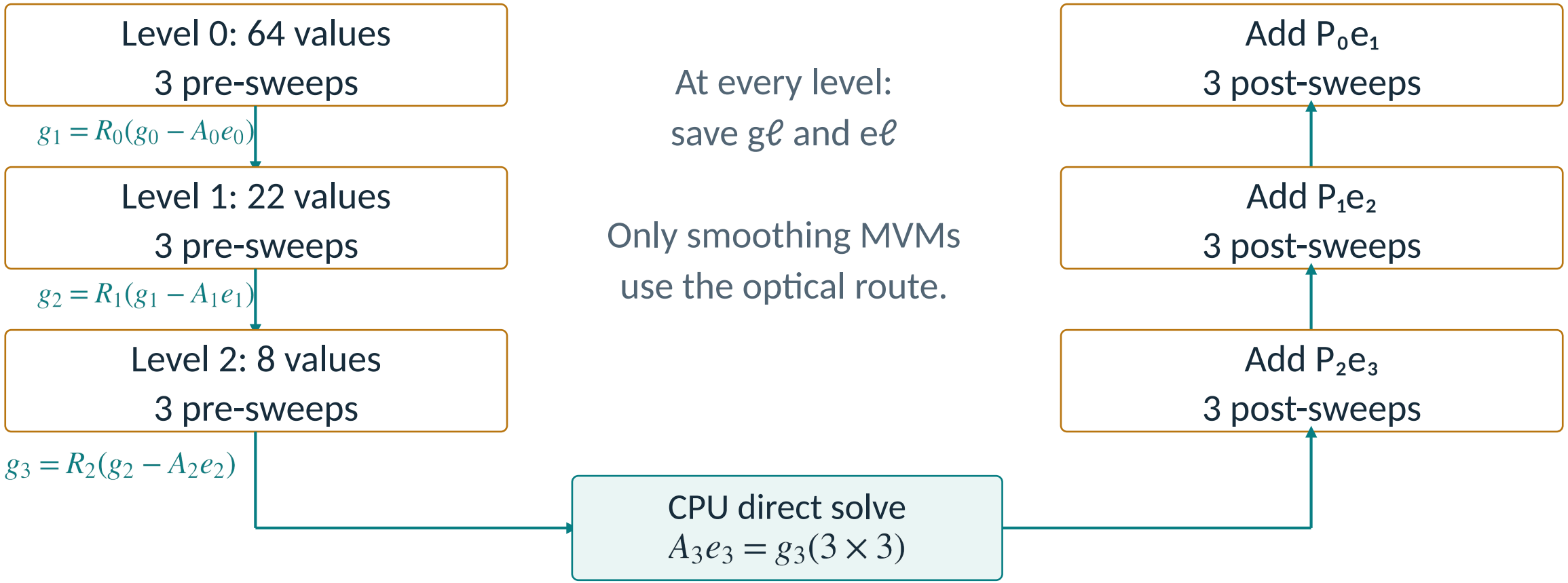
Stored or transferred quantity	Actual first three entries
g_0, retained on CPU	$[4.225e+05, 1, 1, \ldots]^T$
e_0, encoded input	$[25.01, 5.921e-05, 5.921e-05, \ldots]^T$
$y_0 = A_0 e_0$, detected product	$[2.114e+05, -1.057e+05, 0, \ldots]^T$
new $e_0 = e_0 + \omega_0(g_0 - y_0)$	$[37.51, 6.257, 0.0001184, \ldots]^T$

$\omega_0 = 5.9206167e-05$ · 64 entries · full x unchanged

Next sweep encodes the updated e_0 through the same fixed matrix A_0.

The multigrid V-cycle

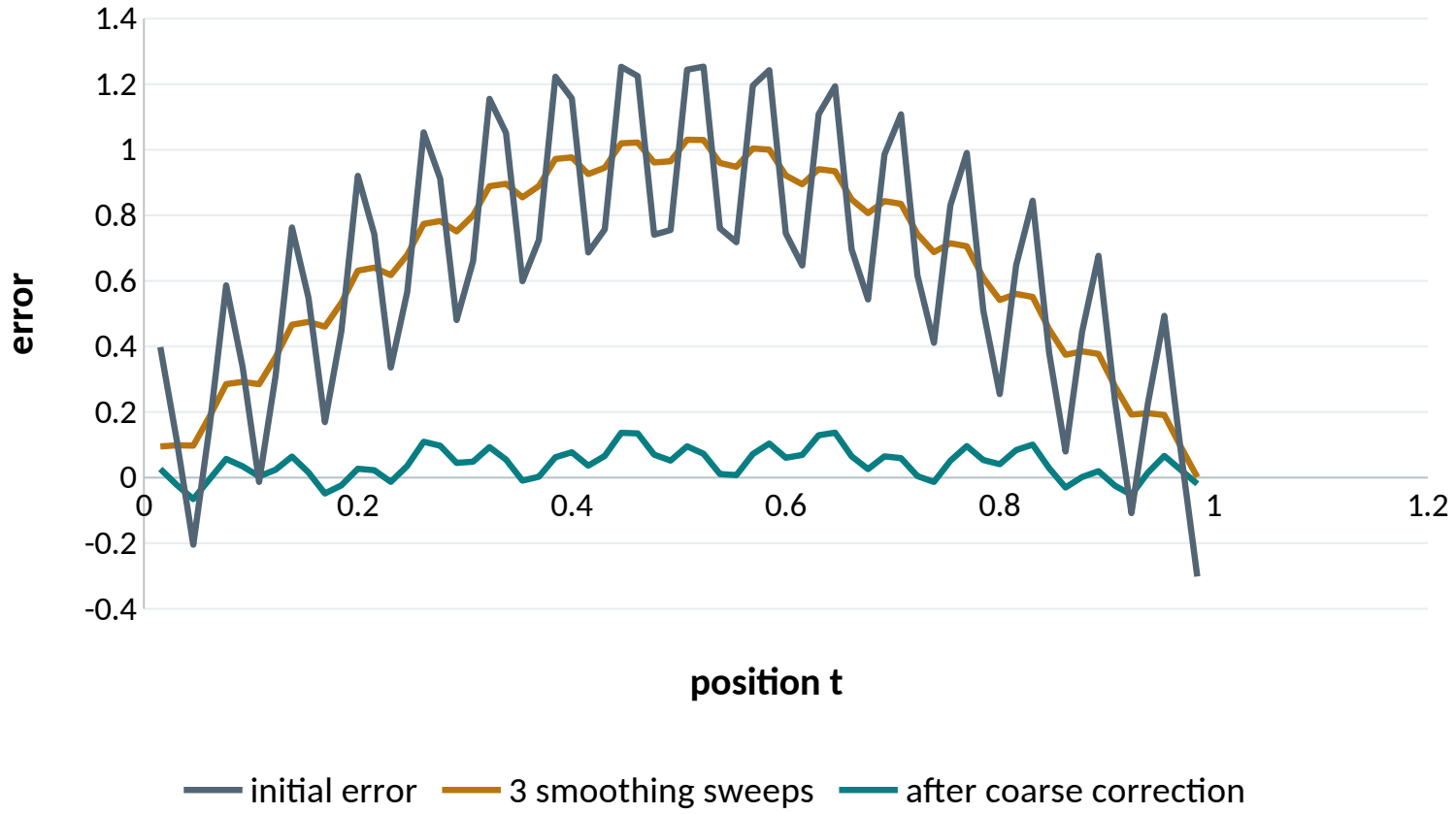
Multigrid V-cycle



Return z = e_0 to PCG. The V-cycle supplies an approximate correction, not the final PDE solution.

Smoothing and coarse correction

Multigrid V-cycle



$$\varepsilon_{\text{new}} = (I - \omega A)\varepsilon$$

Mode j: $1 - \omega\lambda_j$

Error norm

6.04 → 5.70 → 0.50

Three post-sweeps reduce the norm further to 0.422. The initial error is prescribed for this test.

PCG turns a V-cycle correction into a search step

PCG correction

Initialize	$\delta = 0; \quad s = r - A_0\delta$
Precondition	$z = \mathrm{V}(s); \quad p = z; \quad \rho = s^T z$
CPU MVM	$q = A_0 p$
Step length	$\alpha = \frac{\rho}{p^T q}$
Update	$\delta \leftarrow \delta + \alpha p; \quad s \leftarrow s - \alpha q$

p: direction

α: step length

δ: correction so far

PCG arithmetic: CPU

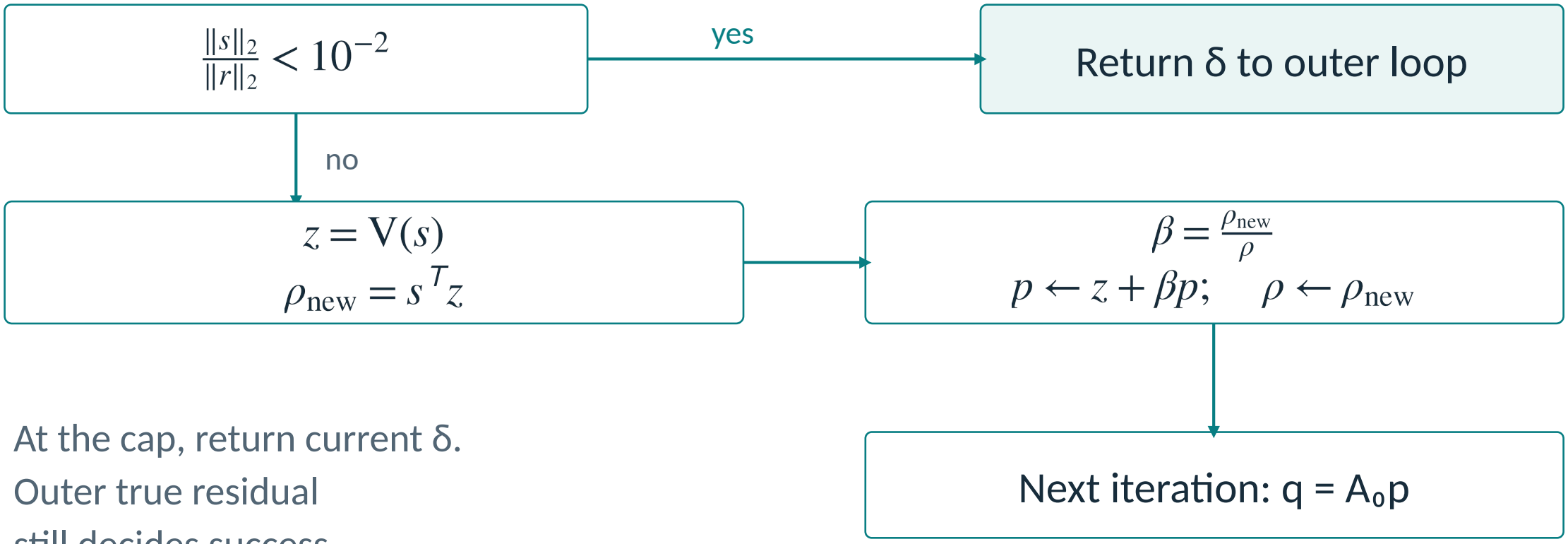
Vcycle invokes

optical products.

The original full estimate x stays fixed until this correction solve returns δ.

PCG stopping and direction updates

PCG correction

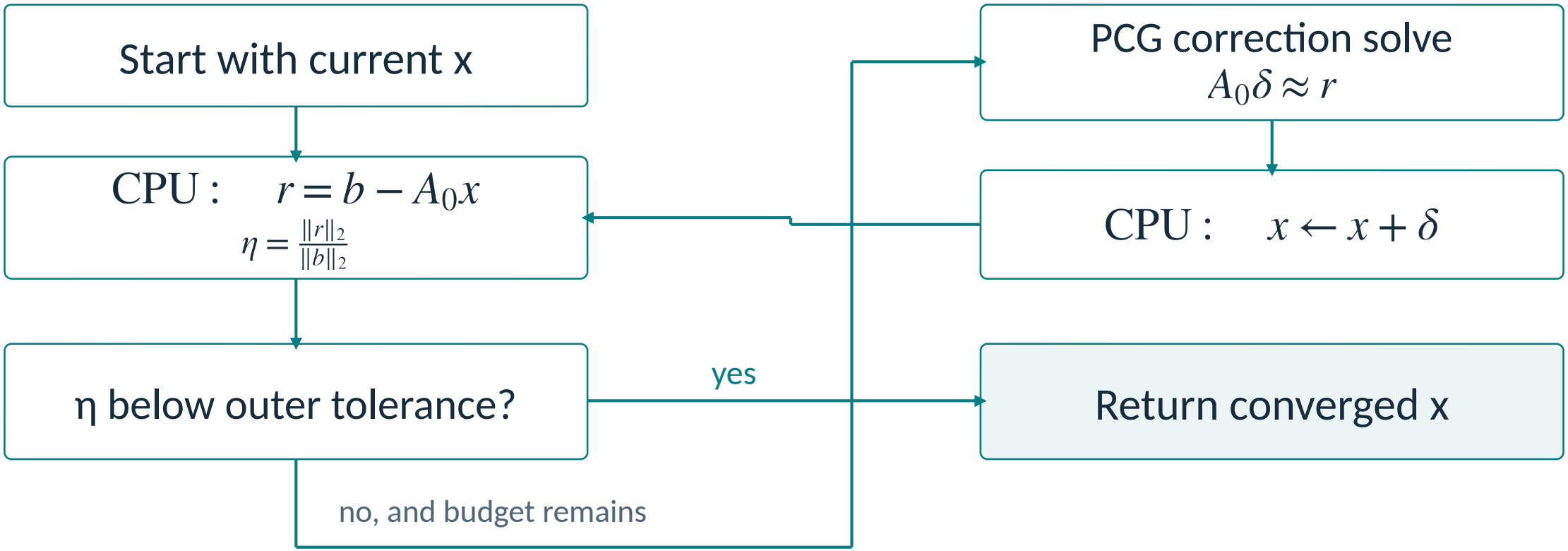


At the cap, return current δ.
Outer true residual
still decides success.

z = Vcycle(s) is an action approximating A_0^{-1}s. No inverse matrix is explicitly stored.

Outer correction and true-residual checks

True residual check



Limit or numerical error: return an unverified approximation with its status, never label it converged.

The hierarchy is fixed while its vectors change

Browser operator inventory: 4 A matrices + 3 P matrices + 3 R matrices

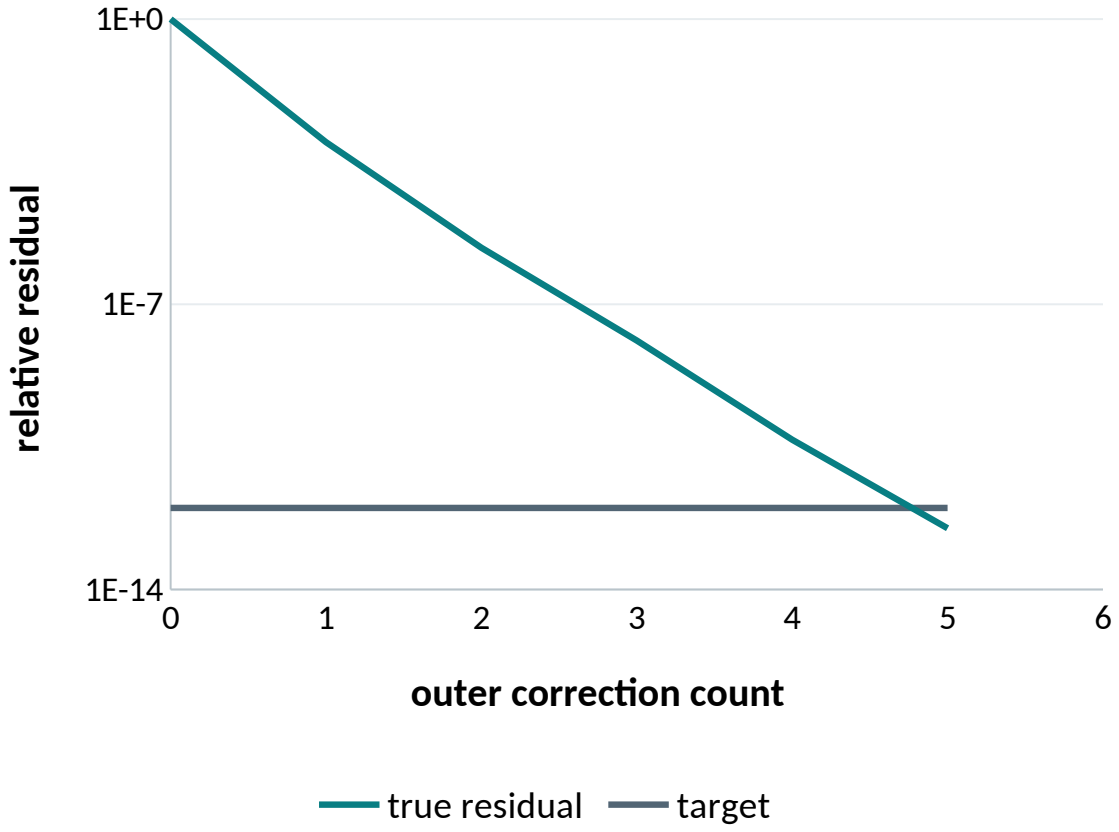
Level	Operator and nonzeros	Prolongation Pℓ	Restriction Rℓ	Assignment
0	A_0: 64 × 64, 190	64 × 22	22 × 64	A: optics + CPU
1	A_1: 22 × 22, 64	22 × 8	8 × 22	A: optics + CPU
2	A_2: 8 × 8, 22	8 × 3	3 × 8	A: optics + CPU
3	A_3: 3 × 3, 7	—	—	CPU direct solve

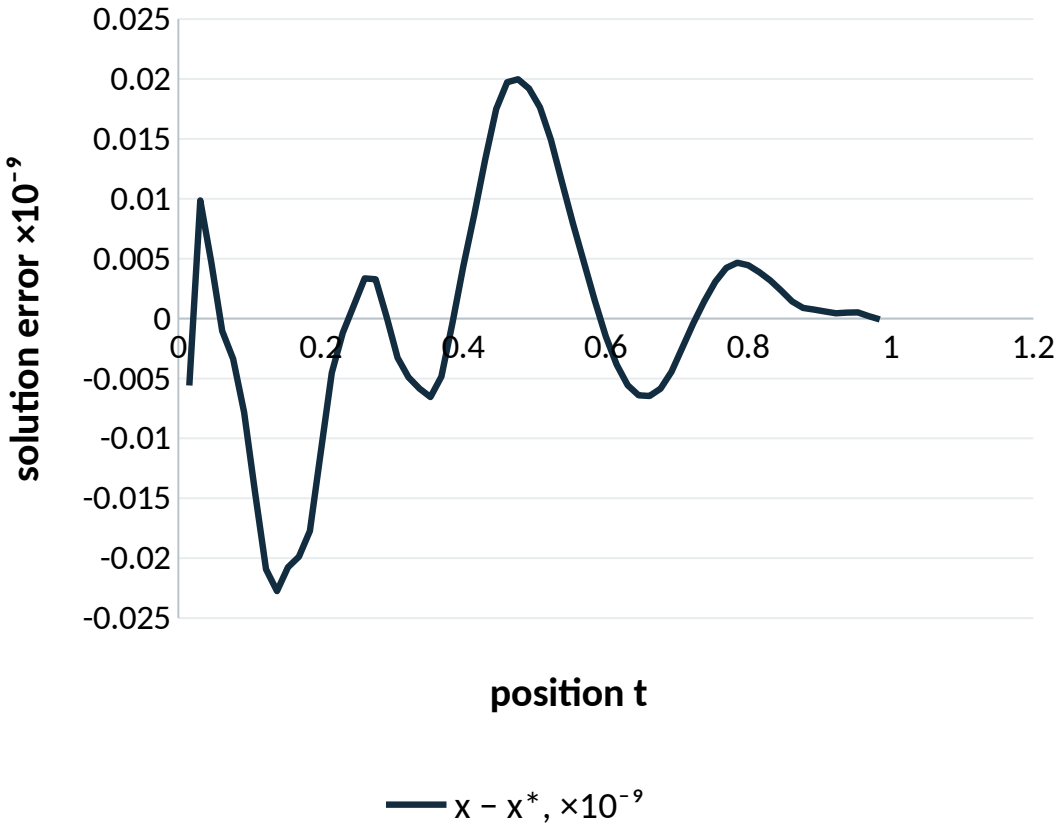
$$A_{\ell+1} = R_\ell A_\ell P_\ell \qquad R_\ell = P_\ell^T \qquad \omega_\ell \approx \frac{1}{\lambda_{\max}(A_\ell)}$$

The CPU retains separate local vectors at each level. Coarse vectors are coefficients in algebraic spaces.

Convergence of the ideal browser model

Ideal 1D model. Zero starting guess. Target 10^{-12}.





Final $\eta = 3.17 \times 10^{-13}$ 180 optical calls 111 digital calls 10 V-cycles

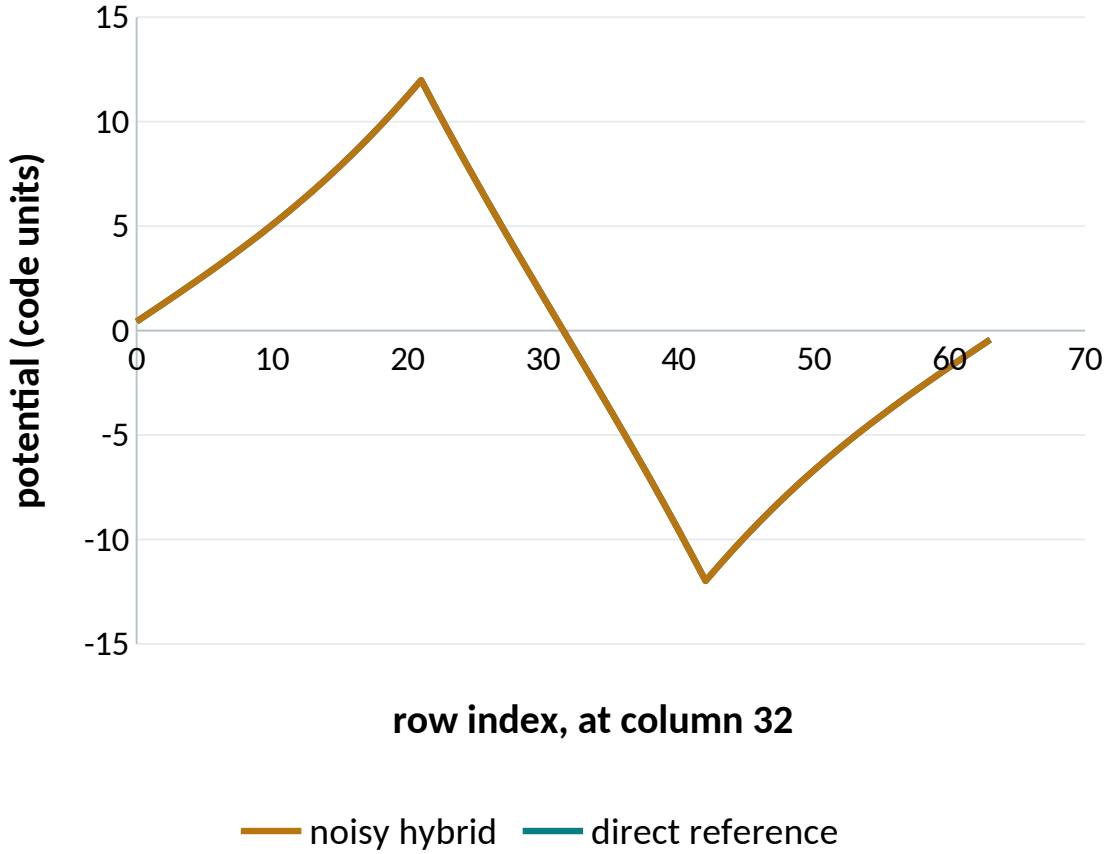
The separate 2D noisy experiment

main.py: 4096 unknowns, two AMG levels, input quantization and output noise

$$4x_{ij} - x_{i-1,j} - x_{i+1,j} - x_{i,j-1} - x_{i,j+1} = b_{ij}$$



64 × 64 interior grid; exterior x = 0



Verified relative residual: 3.96×10^{-11}. Relative solution error: 6.95e-13.

Counted digital products in the 2D run

Same 2D problem and final relative-residual target 10^{-10}. Counter scope matters.

Method	Digital A calls	Analog A calls	True relative residual
Plain CG	148	0	7.8×10^{-11}
Digital MG–CG	60	0	6.1×10^{-11}
Noisy hybrid + restarts	45	60	4.0×10^{-11}

$$148/45 = 3.29$$

Ratio of counted digital products

No measured hardware speedup

Transfers and repeated 704 × 704 coarse solves are excluded. Browser counts use a different convention.

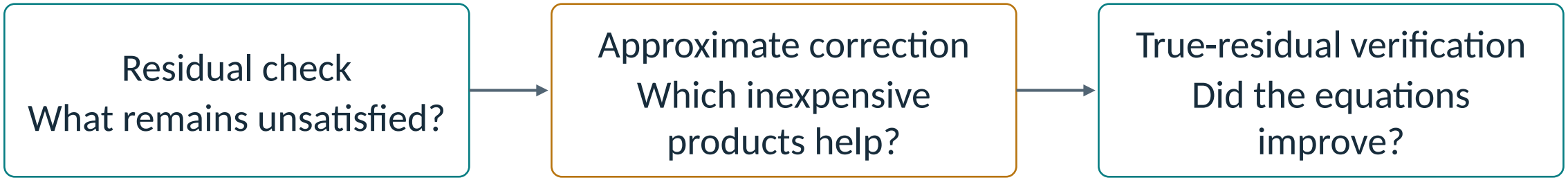
Implemented behavior and open questions

Implemented and checked	Still to establish
Nested residual / CG / multigrid computation	Variable-preconditioner convergence across trials
Input quantization and additive noise in main.py	Calibration against measured optical errors
Fixed matrix blocks and CPU feedback in demonstrator	Signed encoding, sparse tiling, and physical routing
True residuals and algorithmic counters	End-to-end latency and energy, including transfers

Related work: Zhu et al. (2024) provides a coherent signed-MVM architecture with digital memory and control.

Next validation: full computational cost

Optics computes matrix–vector products.
The CPU builds and verifies the solution.



Next: calibrated noise + repeated trials + full cost accounting, with the same sparse operators.

Complete recurrence and termination conditions

Technical appendix: browser configuration

Outer correction

Measure the true residual

$$r = b - A_0 x$$

Check convergence

$$\eta = \frac{\|r\|_2}{\|b\|_2}$$

Approximately solve

$$A_0 \delta \approx r$$

Update the full solution

$$x \leftarrow x + \delta$$

PCG correction

Initialize and precondition

$$\delta = 0; \quad s = r; \quad z = \mathrm{V}(s)$$

Choose direction and step

$$p = z; \quad \rho = s^T z; \quad q = A_0 p$$

Calculate step length

$$\alpha = \frac{\rho}{p^T q}$$

Update correction and residual

$$\delta \leftarrow \delta + \alpha p; \quad s \leftarrow s - \alpha q$$

Outer: 10^{-12} target, 10 corrections. PCG: 10^{-2} relative target, 20 steps; direction update on slide 12.

Recursive V-cycle and smoothing pseudocode

Every operation belongs to a CPU or optical assignment

```
V(l, g):
  if l = 3: return solve(A₃, g)
  e = 0
  repeat 3 times: e = Smooth(l, e, g)
  d = g − Aℓe                    [CPU]
  gc = Rℓd                       [CPU]
  ec = V(l + 1, gc)
  e = e + Pℓec                   [CPU]
  repeat 3 times: e = Smooth(l, e, g)
  return e
```

Smooth(l, e, g):

encode e

y = Aℓe [optical]

detect y

$e = e + \omega_\ell (g - y)$

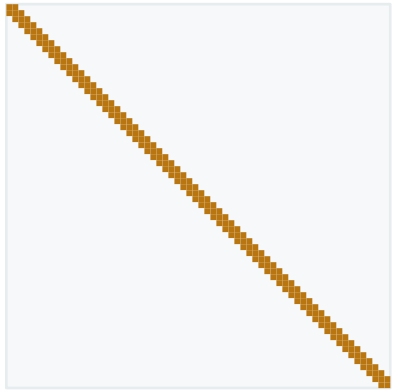
[CPU]

return e

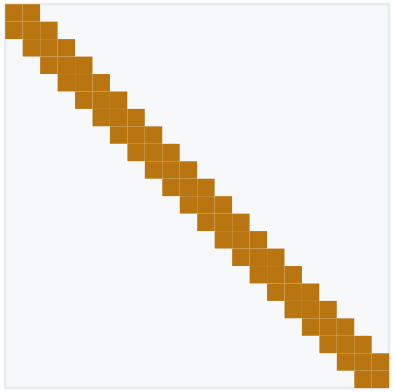
At level 0, g_0 = s and V(0, s) returns z. The outer solution x is separate.

Fine operators A_0 and A_1

A_0: 64 × 64, 190 nonzeros. A_1: 22 × 22, 64 nonzeros.



A_0 fixed
CPU residuals
Optical smoothing



A_1 fixed
CPU residuals
Optical smoothing

i / j	1	2	3	4
1	8450.000	-4225.000	0.000	0.000
2	-4225.000	8450.000	-4225.000	0.000
3	0.000	-4225.000	8450.000	-4225.000
4	0.000	0.000	-4225.000	8450.000

Upper-left 4 × 4 excerpt; rounded to 3 decimals

i / j	1	2	3	4
1	1313.952	-401.563	0.000	0.000
2	-401.563	938.692	-466.913	0.000
3	0.000	-466.913	938.836	-469.236
4	0.000	0.000	-469.236	938.884

Upper-left 4 × 4 excerpt; rounded to 3 decimals

Coarse operators A_2 and A_3

Full matrices shown. Entries rounded to 3 decimals.

i / j	1	2	3	4	5	6	7	8
1	151.590	-25.182	0.000	0.000	0.000	0.000	0.000	0.000
2	-25.182	104.692	-52.098	0.000	0.000	0.000	0.000	0.000
3	0.000	-52.098	104.328	-52.164	0.000	0.000	0.000	0.000
4	0.000	0.000	-52.164	104.328	-52.164	0.000	0.000	0.000
5	0.000	0.000	0.000	-52.164	104.328	-52.164	0.000	0.000
6	0.000	0.000	0.000	0.000	-52.164	104.328	-52.111	0.000
7	0.000	0.000	0.000	0.000	0.000	-52.111	104.642	-25.125
8	0.000	0.000	0.000	0.000	0.000	0.000	-25.125	153.967

A_2 (8 × 8): optical smoothing + digital residuals

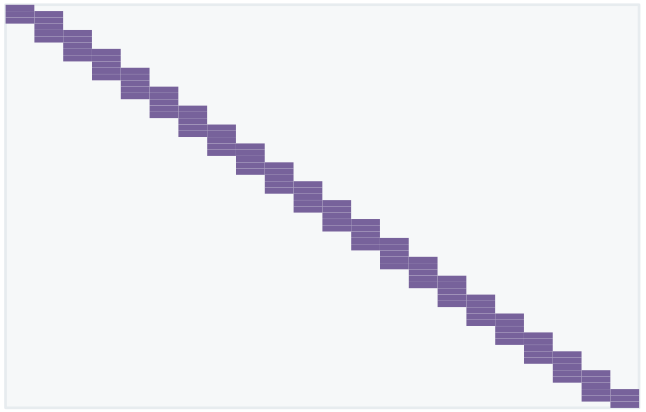
i / j	1	2	3
1	15.416	-1.094	0.000
2	-1.094	11.671	-4.847
3	0.000	-4.847	17.878

A_3 (3 × 3)

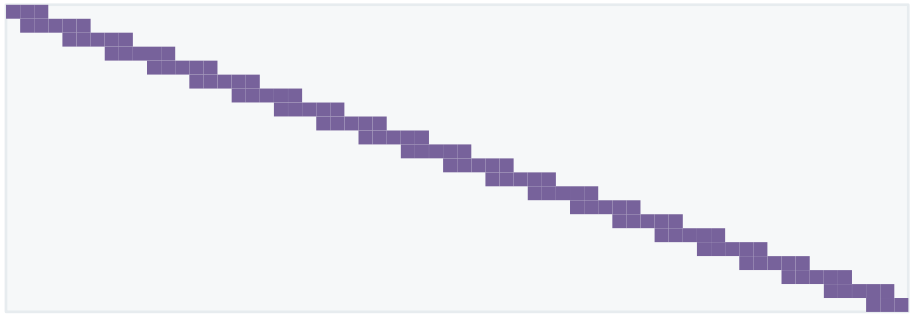
CPU direct solve

Transfer operators P_0 and R_0

P_0: 64 × 22. R_0: 22 × 64. Fixed algebraic transfer operators.



coarse correction to fine



fine residual to coarse: gnext = R d

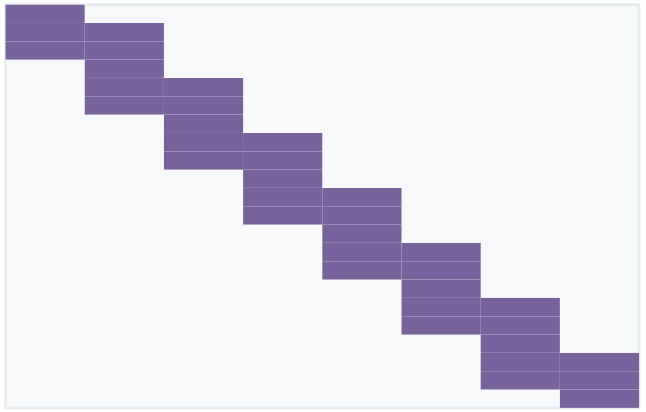
i / j	1	2	3	4
1	0.447	0.000	0.000	0.000
2	0.446	0.156	0.000	0.000
3	0.299	0.349	0.000	0.000
4	0.000	0.572	0.000	0.000
5	0.000	0.415	0.183	0.000

i / j	1	2	3	4	5
1	0.447	0.446	0.299	0.000	0.000
2	0.000	0.156	0.349	0.572	0.415
3	0.000	0.000	0.000	0.000	0.183
4	0.000	0.000	0.000	0.000	0.000

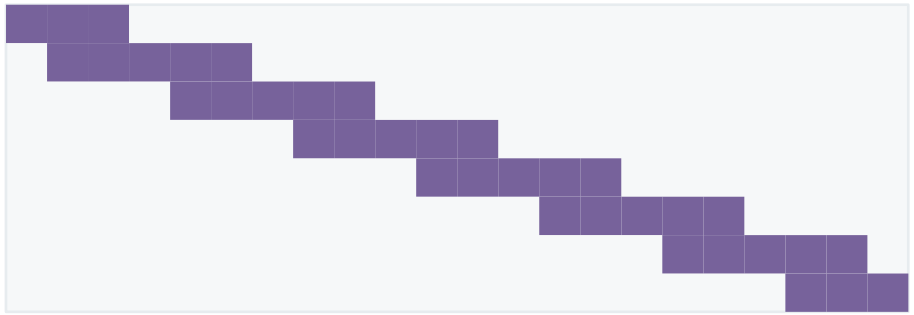
Numerical excerpts: first rows / columns. Full patterns above; zeros outside these excerpts are not implied.

Transfer operators P_1 and R_1

P_1: 22 × 8. R_1: 8 × 22. Fixed algebraic transfer operators.



coarse correction to fine



fine residual to coarse: gnext = R d

i / j	1	2	3	4
1	0.306	0.000	0.000	0.000
2	0.408	0.181	0.000	0.000
3	0.313	0.376	0.000	0.000
4	0.000	0.577	0.000	0.000
5	0.000	0.395	0.193	0.000

i / j	1	2	3	4	5
1	0.306	0.408	0.313	0.000	0.000
2	0.000	0.181	0.376	0.577	0.395
3	0.000	0.000	0.000	0.000	0.193
4	0.000	0.000	0.000	0.000	0.000

Numerical excerpts: first rows / columns. Full patterns above; zeros outside these excerpts are not implied.

Transfer operators P_2 and R_2

Full final transfer pair. $R_2 = P_2^T$. Both operations run on the CPU.

i / j	1	2	3
1	0.2263	0.0000	0.0000
2	0.3424	0.2010	0.0000
3	0.3205	0.3753	0.0000
4	0.0000	0.5773	0.0000
5	0.0000	0.3754	0.2405
6	0.0000	0.2020	0.4377
7	0.0000	0.0000	0.4885
8	0.0000	0.0000	0.1654

i / j	1	2	3	4	5	6	7	8
1	0.2263	0.3424	0.3205	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0000	0.2010	0.3753	0.5773	0.3754	0.2020	0.0000	0.0000
3	0.0000	0.0000	0.0000	0.0000	0.2405	0.4377	0.4885	0.1654

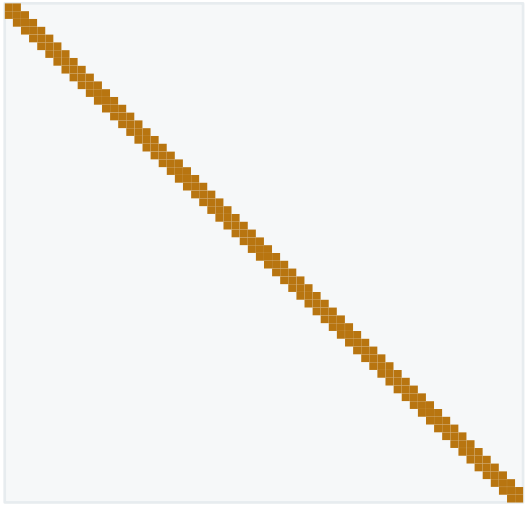
P_2e_3 has 8 entries.

R_2d_2 has 3 entries.

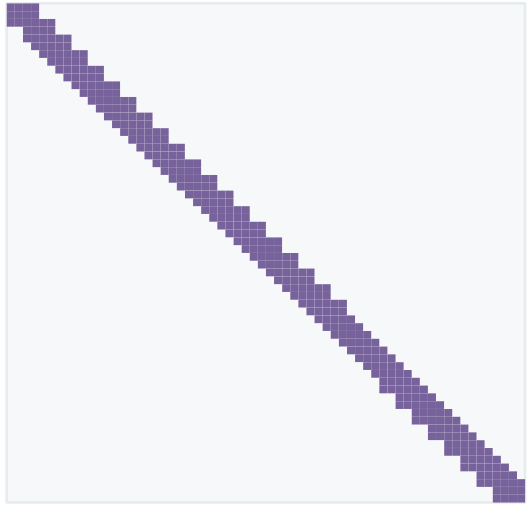
The 2D experiment uses four fixed sparse matrices

main.py hierarchy from the measured run; distinct from the browser matrices

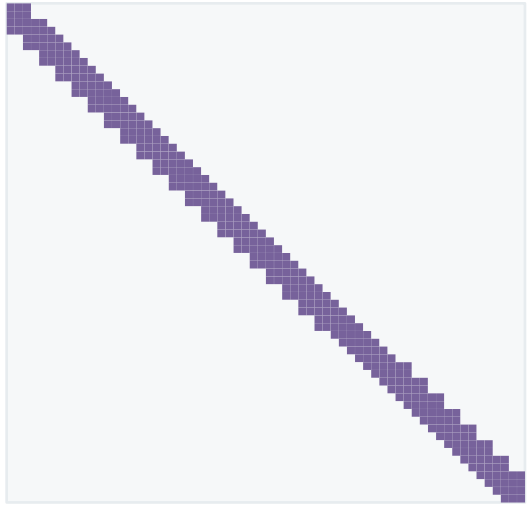
A: 4096 × 4096
20,224 nonzeros



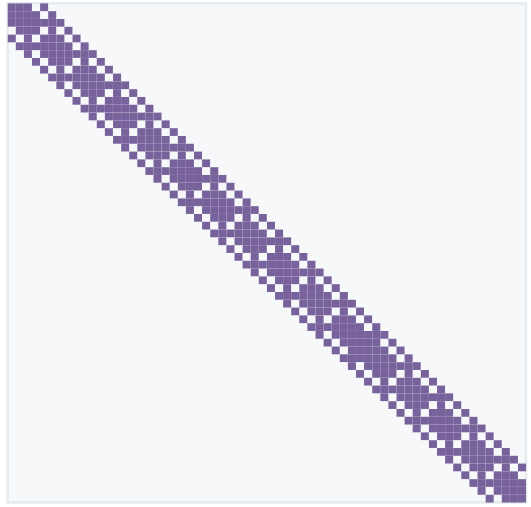
P: 4096 × 704
10,116 nonzeros



R: 704 × 4096
10,116 nonzeros



Ac: 704 × 704
6,096 nonzeros



$$A = I_{64} \otimes T_{64} + T_{64} \otimes I_{64} \qquad Ac = RAP \qquad R = P^{T}$$

T_{64} = tridiag(−1, 2, −1). Patterns use 64 × 64 occupancy bins. Coarse solve: CPU.

Settings and counter definitions

Setting	main.py hybrid	Browser
Outer target / cap	10^{-10} / 100 restarts	10^{-12} / 10 corrections
Inner CG target / cap	10^{-12} / 5 steps	10^{-2} / 20 steps
Smoothing	4 pre / 0 post	3 pre / 3 post
Digital counter	Fine A wrapper calls	$A\ell$, $P\ell$ and $R\ell$ calls
Zero-input smoothing	First product skipped	Explicitly counted
Coarse solve	704 × 704 sparse solve	3 × 3 direct solve
Convergence evidence	Independent true residual	True residual + terminal status

Convergence theory and observed results

Classical PCG assumes a fixed, linear, SPD preconditioner.
Noise and quantization make this V-cycle variable.

$$A(x^\star - x) = r$$
$$x^\star - x = A^{-1} r$$

A small residual measures satisfaction of the discrete equations.
Solution error also depends on conditioning. PDE error also depends on discretization.

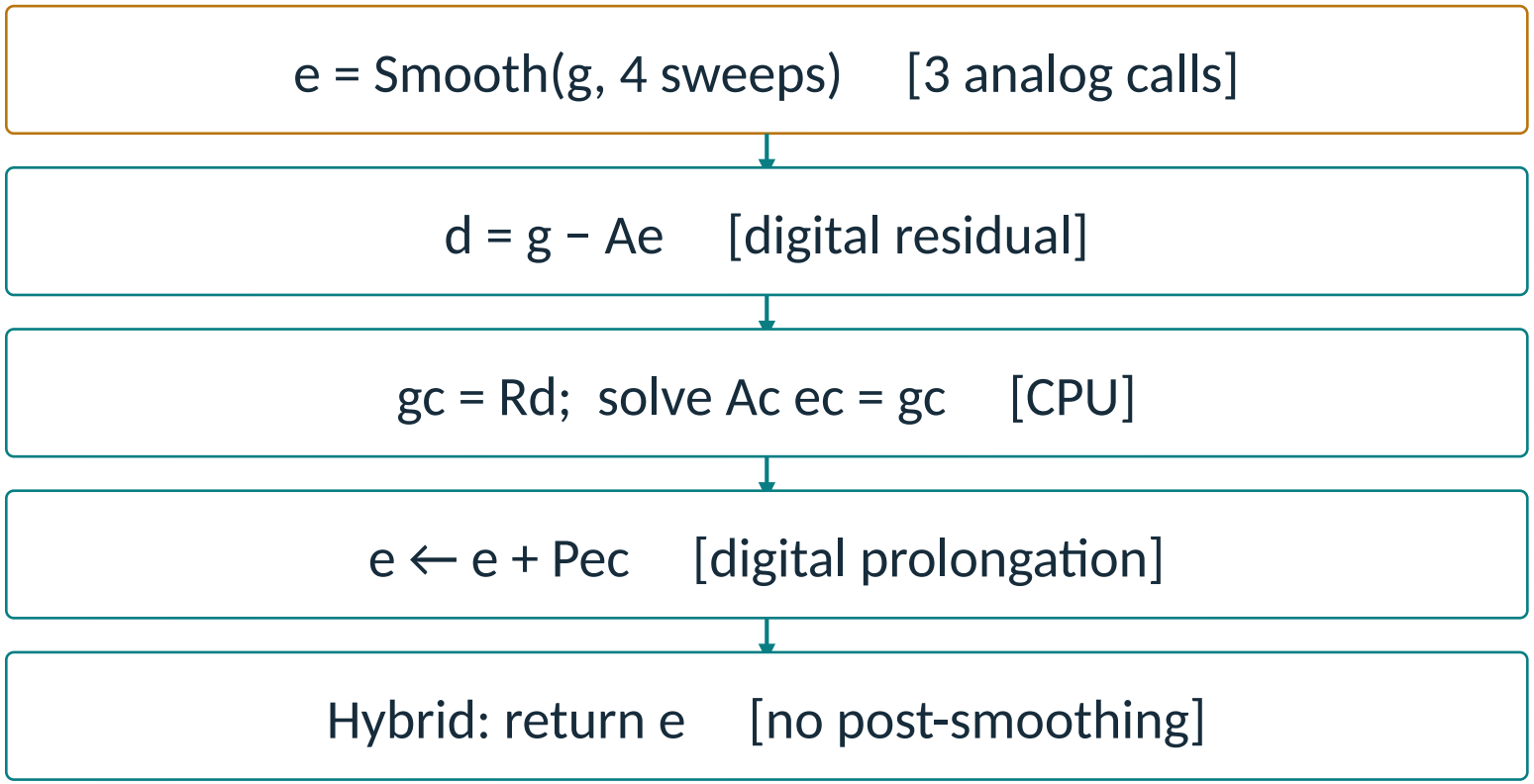
Related work: Zhu’s coherent processor

Aspect	Lee-inspired project	Zhu related work
Numerical emphasis	Multigrid inside Krylov + residual correction	Fixed-point iteration with error management
Optical assignment	Smoothing matrix–vector products	Coherent signed products
Electronics	Residuals, vectors, transfers, control	Memory, reconfiguration, control
Project status	Numerical emulation and ideal demonstrator	Architecture reference; not reproduced

Common principle: approximate optical linear operations cooperate with digital state and correction.

The actual two-level cycle in main.py

4096 → 704. Hybrid: four pre-sweeps, no post-smoothing.



Digital baseline:
2 pre + 2 post

Plain / digital CG:
solve Ax = b

Hybrid only:
short correction solves
with outer restarts

The coarse solver runs on a 704 × 704 matrix each cycle. Its cost is absent from the MVM counters.

Setup constructs operators before iteration starts



Browser level	Stored relaxation weight	Use
0	5.92061673e-05	Richardson smoothing
1	0.000534826731	Richardson smoothing
2	0.0049999113	Richardson smoothing
3	0.0485813754	Unused: direct solve

main.py

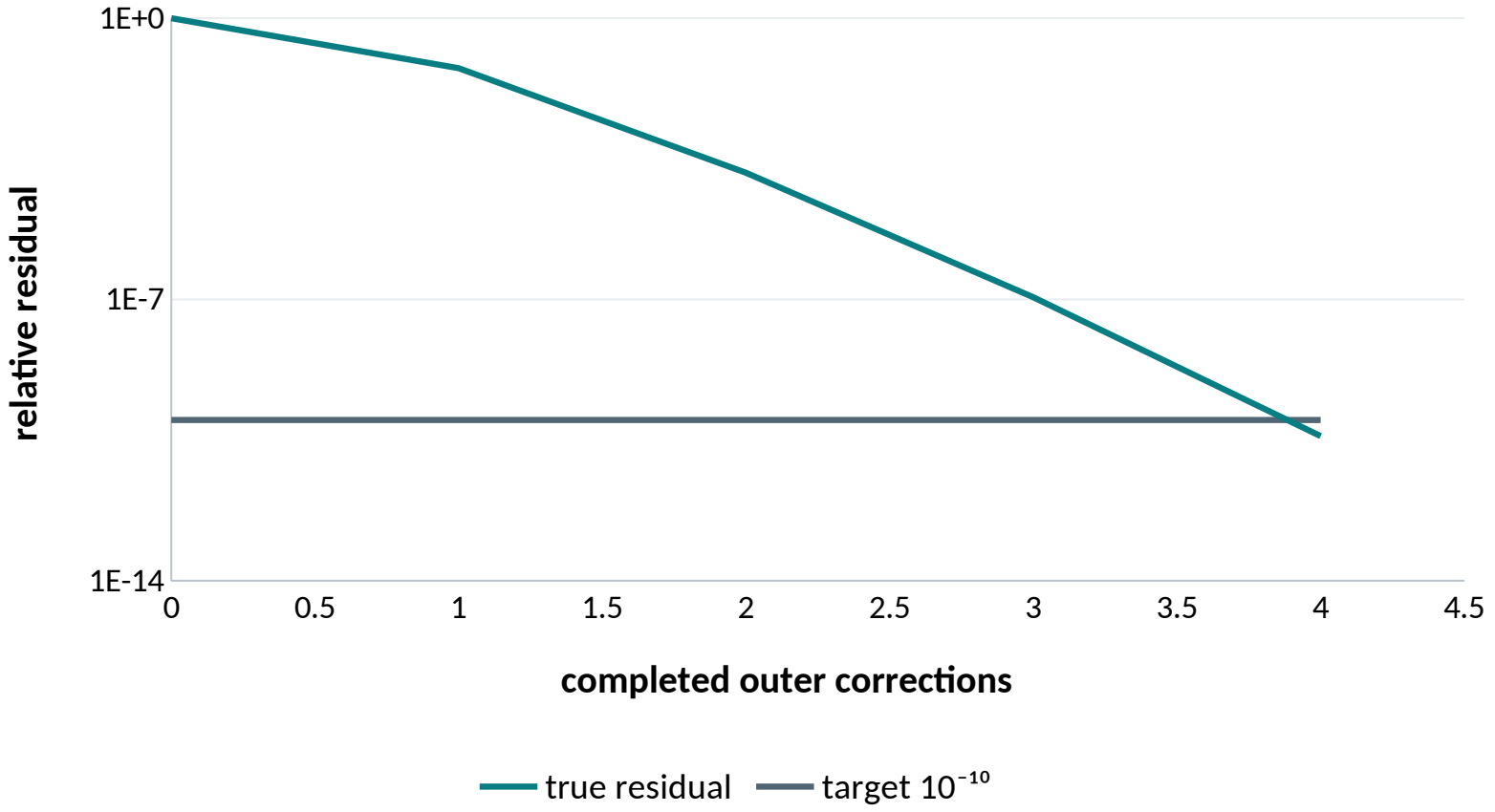
30 power iterations

$$\omega = 0.12717828$$

During the solve: A, P, R, and ω stay fixed. Only solution, residual, and correction vectors change.

True residuals contract across noisy restarts

main.py noisy hybrid. Each correction solve has at most five CG steps.



Four corrections

20 V-cycles

60 analog calls

Observed convergence for this seeded run; robustness across noise realizations remains to be measured.